Hashing
Program

```
#include <stdio.h>
#include<stdlib.h>
#define TABLE_SIZE 10
int h[TABLE_SIZE)=(NULL);

void insert()

{
 int key, index, i, flag=0,hkey;

printf("\nenter a value to insert into hash table\n");
scanf("%d",&key);
 hkey=key%TABLE_SIZE;
for(i=0;i<TABLE_SIZE;i++) { index=(hkey+i)%TABLE_SIZE;
if(h[index] == NULL) {
 h[index]=key;
break;
  }
}
if(i== TABLE_SIZE)
printf("\nelement cannot be inserted\n");

void search()
{
int key, index,
i, flag=0,hkey;
printf("\nenter search element\n");
scanf("%d",&key); hkey=key%TABLE_SIZE;
for(i=0;i<TABLE_SIZE; i++)
{
 index=(hkey+i)%TABLE_SIZE;
if(h[index]==key)
{
 printf("value is found at index %d", index);
}
break;
}
 if(i== TABLE_SIZE)
printf("\n value is not found\n");
}
```

```
void display() {
 int i;
 printf("\nelements in the hash table are \n");
 for(i=0;i< TABLE_SIZE; i++)
 printf("\nat index %d \t value = %d",i,h[i]);
}
int main() {
int opt,i;
while(1)
 {
 printf("\nPress 1. Insert\t 2. Display \t3. Search \t4.Exit \n");
scanf("%d",&opt);
switch(opt) {

case 1:
insert();
break;
case 2:
display();
break;
case 3: search();
break;
case 4:exit(0);
}
}
}
```

OUTPUT:

```
Press 1. Insert 2. Display 3. Search 4.Exit
1
enter a value to insert into hash table
34
Press 1. Insert 2. Display 3. Search 4.Exit
1
enter a value to insert into hash table
 45
Press 1. Insert  2. Display 3. Search  4.Exit
2

elements in the hash table are

at index 0 value = 0
at index 1 value = 0
```

at index 2 value = 0
at index 3 value = 0
at index 4 value = 34
at index 5 value =45
at index 6 value = 0
at index 7 value = 0
at index 8 value = 0
at index 9 value = 0

Press 1. Insert 2. Display 3. Search 4.Exit
3
enter search element
45
value is found at index 5
Press 1. Insert 2. Display 3. Search 4.Exit
4